Output column reference

Every column RSAS can write, what it means, and its units. You won't see all of these in one file; only the columns you enable in the Output Columns dialog get computed and exported. This is the reference for when you're staring at a header and need to know exactly what the number under it represents.

A few conventions used throughout:

- `{T}` stands for a configured temperature in °C. If your run uses 25, 37, and 42, then `Original_MFE_{T}C` expands to three columns: `Original_MFE_25C`, `Original_MFE_37C`, `Original_MFE_42C`.
- **base temperature** is the lowest temperature in your run. It's the reference for hairpin detection and for every difference column.
- **Paired%** always means the percentage of a region's positions that are base-paired in that temperature's MFE structure.
- Composition (AU/GC/GU) is over the *sequence*; pairing and structure columns are over the *fold*.

The columns below are listed in the order RSAS writes them.

Sequence identity

Column	Meaning
`Name`	The sequence name from your input file.
`Sequence`	The input RNA sequence (T converted to U).
`Structure`	MFE dot-bracket structure at the base temperature.
`Full_Structure_{T}C`	MFE dot-bracket at each non-base temperature. The base temperature's structure is the `Structure` column above, so it isn't repeated here.

Full-length energy and composition

Column	Meaning	Units
`Original_MFE_{T}C`	Minimum free energy of the full-length fold at each temperature. More negative = more stable.	kcal/mol
`Original_AU%`	(A + U) / length × 100 for the full sequence.	%
`Original_GC%`	(G + C) / length × 100.	%

Column	Meaning	Units
`Original_GU%`	(G + U) / length × 100.	%

Range checks

These flag whether the value above falls inside the range you configured. Useful for filtering.

Column	Meaning
`Original_MFE_{T}C_InRange`	"In Range" / "Not in Range" for the full-length MFE at each temperature.
`Original_AU%_InRange`	Same, for full-length AU%.
`Original_GC%_InRange`	Same, for full-length GC%.
`Original_GU%_InRange`	Same, for full-length GU%.

RBS sequestering in the full-length fold

How buried the ribosome binding site is when the whole sequence folds, measured at each temperature.

Column	Meaning	Units
`Full_RBS_{T}C_Seq`	The RBS subsequence as identified at each temperature.	—
`Full_RBS_{T}C_Struct`	The RBS region's dot-bracket at each temperature.	—
`Full_RBS_{T}C_Paired%`	Percentage of the RBS that is paired at each temperature. This is the core thermometer reading.	%
`RBS_Seq_Diff_{hi}-{base}`	Difference in RBS paired% between the highest temperature and the base temperature. A large negative value (paired when cold, open when warm) is the thermometer fingerprint.	percentage points
`RBS_Seq_Diff_{mid}-{base}`	Same difference for the middle temperature versus base (only when you have 3+ temperatures).	percentage points

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Hairpin analysis

The extracted regulatory hairpin: how it was found, its sequence and structure, and its own energy and composition.

Column	Meaning	Units
`Hairpin_Detection_Method`	Which method found this hairpin (terminal or RBS-based).	—
`RBS_Detection_Params`	The anchor / side / window used for RBS detection this run, e.g. `AUG/last/5-13`. Records your settings alongside the result.	—
`Hairpin_Sequence`	The extracted hairpin subsequence.	—
`Hairpin_Structure`	The hairpin's dot-bracket structure.	—
`Hairpin_AU%` / `Hairpin_GC%` / `Hairpin_GU%`	Base-**pair** composition of the hairpin: the fraction of its folded base pairs that are A-U, G-C, and G-U wobble. These are over pairs, not bases, so they sum to 100. (Unlike the `Original_*` columns above, which are single-nucleotide frequencies.)	% of pairs
`Hairpin_MFE_{T}C`	MFE of the hairpin folded on its own at each temperature.	kcal/mol

Hairpin range checks

Column	Meaning
`Hairpin_MFE_{T}C_InRange`	Whether the hairpin MFE is in range, per temperature.
`Hairpin_AU%_InRange` / `Hairpin_GC%_InRange` / `Hairpin_GU%_InRange`	Whether hairpin composition is in range.

RBS (standalone)

The RBS as a region in its own right, separate from the full-length sequestering columns above.

Column	Meaning	Units
`RBS_Sequence`	The identified Shine-Dalgarno / RBS subsequence.	—
`RBS_Structure`	Its dot-bracket.	—
`RBS_Paired%`	Percentage of the RBS that is paired.	%
`RBS_Paired%_InRange`	Whether RBS paired% is in your configured range.	—

Partition function

Ensemble-level numbers, only present when you enable PF (it's the slow part of a run). Unlike the MFE columns, these account for all competing structures, not just the single best one.

Column	Meaning	Units
`PF_Full_Ensemble_{T}C`	Ensemble free energy of the full-length sequence at each temperature. At or below the MFE.	kcal/mol
`PF_Full_MeanPaired_{T}C`	Mean pairing probability across the full sequence.	probability (0–1)
`PF_HP_Ensemble_{T}C`	Ensemble free energy of the hairpin.	kcal/mol
`PF_HP_MeanPaired_{T}C`	Mean pairing probability across the hairpin.	probability (0–1)
`PF_RBS_Access_{T}C`	RBS accessibility: mean probability that the RBS positions are unpaired across the ensemble, as a percentage. Higher = more accessible to the ribosome.	% (0–100)
`PF_RBS_Diff_{hi}-{base}`	Change in RBS accessibility from base to highest temperature. The ensemble-level version of the thermometer signal.	percentage points
`PF_RBS_Diff_{mid}-{base}`	Same for the middle temperature (3+ temperatures only).	percentage points
`PF_HP_Ensemble_{T}C_InRange`	Whether the hairpin ensemble energy is in range, per temperature.	—

Motif / sequence finder

Present only when motif search is enabled. When a sequence has several matches, the multi-value fields are semicolon-separated in the CSV, and the Excel file gets a dedicated Motif Matches tab with one row per hit.

Column	Meaning	Units
`Motif_Pattern`	The IUPAC pattern you searched for.	—
`Motif_Count`	Number of matches found in the sequence.	count
`Motif_Match_Seq`	The matched subsequences.	—
`Motif_Match_Pos`	Match positions, 0-based and half-open.	—
`Motif_Paired%_{T}C`	Paired percentage of each match at each temperature.	%
`Motif_Struct_{T}C`	Each match's dot-bracket at each temperature.	—

Column	Meaning	Units
`Motif_PF_Access_{T}C`	Partition-function accessibility of each match: mean unpaired probability over the match, as a percentage (needs PF on).	% (0–100)
`Motif_Paired_Diff_{hi}-{base}`	Change in motif paired% from base to highest temperature.	percentage points
`Motif_PF_Diff_{hi}-{base}`	Change in motif PF accessibility, base to highest.	probability points
`Motif_Paired_Diff_{mid}-{base}` / `Motif_PF_Diff_{mid}-{base}`	Same differences for the middle temperature (3+ temperatures).	—

Quality scores

Two parallel sets: `HP_` scores the hairpin, `FL_` scores the full-length sequence. Both are driven by the profile you set in the Quality Score Builder. See methods.md for exactly how they're computed.

Column	Meaning
`HP_Quality_Score` / `FL_Quality_Score`	Raw score as `passed/evaluated`, e.g. `5/6`. "Passed" means a criterion scored a full 1.0.
`HP_Quality_Score_Weighted` / `FL_Quality_Score_Weighted`	Weight-weighted average of the criterion scores, as a percentage (0–100).
`HP_Quality_Score_Class` / `FL_Quality_Score_Class`	Tier from the weighted percentage: Tier 1 (≥83%), Tier 2 (≥67%), Tier 3 (≥50%), Tier 4 (≥33%), Tier 5 (below).
`HP_Quality_Score_Breakdown` / `FL_Quality_Score_Breakdown`	Per-criterion scores, semicolon-separated, so you can see which criteria carried or sank the total.

Results are sorted by quality score, best first, so your strongest candidates are already at the top of the export.